

EFFECT OF HANG CLEANS OR SQUATS PAIRED WITH COUNTERMOVEMENT VERTICAL JUMPS ON VERTICAL DISPLACEMENT

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ABSTRACT

Andrews, TR, Mackey, TA, Inkrott, TA, Murray, SR, Clark, IE, and Pettitt, RW. Effect of hang cleans or squats paired with countermovement vertical jumps on vertical displacement. *J Strength Cond Res* 25(9): 2448–2452, 2011—Complex training is characterized by pairing resistance exercise with plyometric exercise to exploit the postactivation potentiation (PAP) phenomenon, thereby promising a better training effect. Studies on PAP as measured by human power performances are equivocal. One issue may be the lack of analyses across multiple sets of paired exercises, a common practice used by athletes. We evaluated countermovement vertical jump (CMJ) performance in 19 women, collegiate athletes in 3 of the following trials: (a) CMJs-only, where 1 set of CMJs served as a conditioning exercise, (b) heavy-load, back squats paired with CMJs, and (c) hang cleans paired with CMJs. The CMJ vertical displacement (3-attempt average), as measured with digital video, served as the dependent variable of CMJ performance. Across 3 sets of paired-exercise regimens, CMJ-only depreciated 1.6 cm and CMJ paired with back squats depreciated 2.0 cm (main effect, $p < 0.05$). Conversely, CMJ paired with hang cleans depreciated 0.30 cm (interaction, $p < 0.05$). Thus, the best complex training scheme was achieved by pairing CMJs with hang cleans in comparison to back squats or CMJs in and of themselves. Future research on exercise modes of complex training that best help athletes preserve and train with the highest power possible, in a given training session, is warranted.

KEY WORDS complex training, paired exercise, postactivation potentiation, weightlifting

INTRODUCTION

Complex training is characterized by pairing repetitions of a resistance exercise shortly preceding engaging in a series of biomechanically similar plyometric actions (5,13). Most research on this topic has emerged within the past decade (5), but complex training can be traced back to the mid-20th century (7,17).

The rationale for complex training is that it can elicit an increase in muscular power that may not be recognized by performing the plyometric exercise separately. Complex training theory implies, specifically, that a conditioning exercise such as a squat paired with a similar plyometric action, such as a countermovement vertical jump (CMJ), will evoke a phenomenon termed postactivation potentiation (PAP) (13). By engaging in such paired exercises, augmentation of plyometric exercises from PAP theoretically leads to superior power adaptations (5). Enhanced myosin regulatory light chain phosphorylation and Ca^{2+} sensitivity have been purported as mediating factors for PAP (12); however, other mechanisms may exist. In isolated muscle preparations, the PAP phenomenon is observed consistently (i.e., greater peak tension and shorter time to peak tension); however, such an acute augmentation of power is inconsistent across human studies (5,14).

The speed of the resistance exercise preceding execution of a subsequent set of plyometric exercises may impact the magnitude of power augmentation. Higher speed motions recruit type 2 motor units (4) and such fibers reportedly phosphorylate myosin ATP more readily (10). Indeed, several investigators (2,8,16) have suggested that power-trained athletes are better capable of experiencing acute, transient power augmentation on a consistent basis. Thus, it would seem plausible that pairing a high velocity, weightlifting technique, in comparison to a heavy-load conditioning exercise, would evoke better plyometric performance. McCann and Flanagan (11) examined this hypothesis and reported that the hang clean compared to the heavy-load back squat, used as a preparatory exercise, produced an equivalent effect on subsequent CMJs; the effect on multiple sets of exercise was not explored.

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A baseline power test, not preceded by a conditioning exercise, is generally used as a reference measure to study PAP in humans (5). The issue with such a methodological approach is that the baseline power test may, itself, act to potentiate and be a source of fatigue for a subsequent plyometric action. Repeated twitches in isolated muscle demonstrate potentiation followed by gradual repetition-dependent decrements in tension (15). Similarly, across multiple sets of complex training sessions, power during the plyometric performances hypothetically decreases. At present, we have not located any study on multiple sets of paired exercises that take into account depreciation of performance that may occur normally across sets of voluntary exercise. Lack of research on multiple sets is troubling given that athletes are routinely advised to engage in multiple set vs. single set exercise regiments (9). Therefore, we examined the following 2 research questions. Is a weightlifting exercise superior to heavy-load resistance exercise paired with a biomechanically similar plyometric action? Is complex training worse, better, or equivalent to engaging in the plyometric exercise alone?

METHODS

Experimental Approach to the Problem

This study used a repeated measure design consisting of 3 visits. In counterbalanced order, subjects completed 3×4 reps per sets of CMJs-only (control trials), 3×3 back squats paired with CMJs, and 3×3 hang cleans paired with CMJs. For the CMJ-only trials, an initial set of CMJs served as the conditioning exercise. Vertical displacement for the CMJs served as the dependent variable. All tests were conducted during the same time of the day and within a 2-week period among each subject.

The rationale for the study was as follows: Both CMJ itself (3) and responses of PAP (11,18) are purported to have individual variation, making observation of the PAP phenomena difficult. To control for this, we report consistency measures of vertical displacements across 3 repetitions for 3 experimental trials. Moreover, as complex training is often performed with multiple sets (9), and PAP is largely related with fatigue, we examined CMJ performance in response to 3 sets of exercise. In short, the experimental design allowed for comparing multiple sets CMJs-only vs. complex pairing of with a weightlifting exercise and a heavy-load resistance exercise.

Subjects

A total of 19 women, collegiate athletes (age 20.5 ± 1.5 years, height 172.1 ± 6.0 cm, weight 70.2 ± 8.3 kg) volunteered for this study. Represented sports included basketball (Division II), hockey (Division I), soccer (Division II), and volleyball (Division II). The sample was homogeneous on their maximum performances for the back squat (1.49 ± 0.30 body weight [BW]), hang clean (1.01 ± 0.15 BW), and CMJ (54.4 ± 6.8 cm, 3-trial averaged value). The group was trained previously on the CMJ technique (arm swing) and used

complex training as a regular part of their preexisting training regime. Each athlete was in the off-season. Written informed consent was gathered before commencing data collection. All procedures for this study were preapproved by the host institutional review board.

Procedures

A standard barbell (20 kg) and weight plates (nearest 2.5 kg) were used for 1-repetition maximum (RM) testing and for the back squat and hang clean exercises. Actual 1RM tests were conducted using a specific set of testing standards described elsewhere (1). Before data collection, subjects engaged in their normal routine of warm-up calisthenics (e.g., high knees, carioca steps). The subjects' shirts were then tied off above the waist, and their shorts were pulled down sufficient to expose the iliac crest adequately. A reflective marker was placed over the skin on the most superior point of the right iliac crest. This marker served as surrogate location of center of mass for the body.

A digital video camera (Flip Video, Pure Digital Technologies) operating at 30 Hz was set on a leveled tripod 10 m from the subject. The CMJs (resetting before each jump) were digitized manually with commercial software (Dartfish 5.0, Colorado Springs, CO, USA) to determine vertical displacement (note: automated digitizing was not used). The 3-trial average values were recorded as the dependent variable. For the paired trials, an intensity approximating 75% 1RM of the back squat and 60% for the hang clean were used. Such intensities are representative intensities for the power phase of a periodized resistance exercise program and comparable to what would be used with complex training (9). Each set consisted of 3 repetitions followed by 3 minutes of recovery, then performance of the 3 CMJs.

Baseline CMJs were not gathered before the initial set of back squats or hang cleans. Such a procedure was to avoid having an initial set of CMJs become part of the regimen. Thus to monitor strength between days of testing, we opted to rely on measures of maximum isometric, right knee extension strength (to the nearest 0.1 kg). In a seated position with knee flexed, as verified with a mechanical goniometer, subjects were instructed to extend their right knee thereby pressing the toe of their right shoe into a mechanical transducer (Lafayette Manual Muscle Test System, Lafayette, IN, USA). The transducer was positioned against a cement wall to provide a consistent isometric measurement.

Statistical Analyses

All summary statistics are reported as mean $\pm$ SD. Relative reliability of the vertical displacements for the CMJs was calculated using intraclass correlation coefficients, whereas absolute reliability was determined using typical error and coefficient of variation (CV) (6). Potential differences between conditions (i.e., CMJ-only, back squat, and hang clean) between sets of CMJs were evaluated using repeated-

measures analysis of variance (ANOVA) in 2 separate ways. Firstly, the personal best set followed by 2 subsequent sets of CMJs-only vs. CMJs subsequent to sets 1–3 of the back squats and hang cleans were compared. Secondly, sets 1–3 of the CMJs-only, after an initial set of CMJs as the conditioning exercise, were compared with sets 1–3 of the back squats and hang cleans. Significant interaction was evaluated using the Bonferroni adjustment. One-way ANOVA with repeated measures was used to evaluate for differences in the open kinetic knee extension test.

RESULTS

Vertical displacement values for the 3 attempts of the CMJs per set remained consistent between trials (Table 1). There was a tendency for more consistent vertical displacements for the day that hang cleans were paired with CMJs. Measures on the open kinetic chain knee extension test did not differ ($F = 0.39$, $p > 0.68$, $\eta^2 = 0.07$) indicating knee extension strength differences, per day, did not serve as a confounding variable for this study.

For the CMJ-only trial, vertical displacement depreciated 1.6 cm across sets. Vertical displacement of CMJ-only (inclusive of the personal best set) were not significantly different from CMJs that followed either the back squat or hang clean (interaction: $F = 1.71$, $p = 0.15$, $\eta^2 = 0.06$). Similar to CMJ-only, vertical displacement for CMJ subsequent to the back squat and hang clean also depreciated (main effect: $F = 5.37$, $p < 0.05$, $\eta^2 = 0.19$). When the fatiguing effect of an initial set of 3 CMJs was considered, relative to set number, vertical displacement depreciated more for the CMJ-only and back squat in comparison to the hang clean (interaction: $F = 2.42$, $p = 0.04$, $\eta^2 = 0.39$). Thus, the athletes were better able to maintain CMJs performance when paired

TABLE 1. Reliability analyses for vertical jumping measurements.*

		ICC alpha	Typical error (cm)	CV (%)
CMJ-only	Personal best	0.81	3.06	5.8
	Set 1	0.77	3.25	6.2
	Set 2	0.69	3.28	6.6
	Set 3	0.81	2.71	5.7
Hang clean	Set 1	0.83	2.44	4.9
	Set 2	0.77	3.10	6.2
	Set 3	0.80	2.76	5.3
Back squat	Set 1	0.74	3.47	7.1
	Set 2	0.88	2.24	4.6
	Set 3	0.84	2.55	5.2
Average		0.79	2.89	5.8

*ICC = intraclass correlation; CMJ = countermovement vertical jump; CV = coefficient of variation.

TABLE 2. Vertical displacement (cm) for the 3 conditions (data based on average of 3 trials per set).*

Personal best	54.4 ± 6.2	
Set 1	Set 2	Set 3†‡
CMJ-only		
53.4 ± 7.1	52.8 ± 6.2	51.5 ± 6.0
Set 1	Set 2	Set 3†‡
Back squat		
53.7 ± 6.2	51.9 ± 6.1	51.7 ± 6.0
Set 1	Set 2	Set 3†
Hang clean		
52.4 ± 5.5	52.6 ± 5.8	52.1 ± 5.9

*Data reported as mean ± SD; CMJ = countermovement vertical jump.

†Main effect ($p < 0.05$) decrease in vertical displacement relative to set 1.

‡Interaction ($p < 0.05$) noting a significant between-trial difference of the hang clean vs. the CMJ-only and the back squat in the third set.

with the hang clean vs. the back squat paired with CMJs or performing CMJs exclusively (Table 2).

DISCUSSION

The first research question for this study was to compare a fast speed conditioning exercise (hang clean) and a heavy-load resistance exercise (back squat) paired with CMJs. Our findings indicate that the hang clean was superior to the back squat at maintaining consistent CMJ performance (Table 2). As such, we support the hypothesis that a fast speed, weightlifting exercise paired with a plyometric action is superior to using slow, heavy-load exercise paired with a plyometric action.

The second research question for this study was to examine if a complex training approach was worse, better, or equivalent to simply performing the plyometric exercise in it of itself (i.e., CMJ-only). We compared trials pairing repetitions of the back squat and the hang clean with CMJs vs. and a trial involving CMJs-only. Thus, instead of pairing with resistance exercise, a set of CMJs served as the conditioning exercise. We observed that hang cleans were superior in maintaining consistent CMJs performance, whereas CMJs in of themselves depreciated to a greater extent. Such a finding supports the hypothesis that complex training offers an advantage to execution of plyometric exercise exclusively.

Individual variation has been indicated as a confounding variable explaining why many investigators are unable to observe PAP for a group of athletes (11,18). McCann and Flanagan (11), for instance, observed no differences between the back squat and hang clean on potentiating a single set of CMJs on a sample of volleyball athletes. Based on

a surveillance of their individual data; however, the authors suggested optimum rest between the conditioning exercises and CMJs was “highly individualized.” Similarly, using the squat jump with force platform measurements, Weber et al. (18) evaluated a group of track athletes (i.e., homogeneous in their muscular power and training status). These investigators observed several subjects who potentiated, whereas others depreciated in their performance. They concluded that no single complex training regiment may be optimum because of individual variability.

Although the magnitude of PAP (i.e., amplitude of twitch force) on a single twitch of an isolated muscle fiber is pronounced (14), the magnitude of PAP on voluntary plyometric actions in humans is typically 2–4% (13). Assessment of CMJ vertical displacement can impose a barrier for recognizing PAP in that the CV for this measurement is typically 5%, although <10% variability has been deemed reliable for CMJ analysis (3). We observed CV values of ~5.8% (Table 1). Such a CV value is far better than a mechanical measure of vertical displacement (e.g., Vertec device) and comparable to vertical displacements based on flight times using a force platform. With strong reliability data, we are confident in our ability to detect individual differences among athletes and changes in vertical displacement between conditions; however, we concede more precise and valid values would be gained from better sampling rates to the video recorder we used.

Cormack et al. (3) suggested computation of a smallest clinically worthwhile change (SWC) value may be useful in detecting finite changes in performance on the CMJ. In brief, the value of 0.2 multiplied by the between-subject *SD*, which is rooted in Cohen’s effect size guideline for minimal perceptible difference of 20%, was applied in their study of CMJ reliability. These authors indicated that a 0.7-cm difference in vertical displacement is needed to detect SWC in CMJ performance; however, as we used cruder technology, our SWC value was ~1 cm. Thus, when comparing how much CMJ vertical displacement values depreciated to the final set (Table 2), depreciation of the vertical displacement in the hang clean did not exceed our SWC. Conversely, group values of the CMJ vertical displacements depreciated >1 cm for the back squat and CMJ-alone conditions. Thus, using the SWC criteria, our findings are both significant and clinically worthwhile.

We confined our study to a convenient sample of women athletes; thus, whether our results extend to other populations is speculative. The single dependent variable of vertical displacement was used and no direct data on muscular activation was gathered. More sensitive measures on CMJ with lower typical errors values are available (3), along with techniques for studying neuromuscular responses (e.g., electromyography). Thus, our findings on multiple sets should be viewed as preliminary findings showing promise that pairing weightlifting with plyometric

actions may be more advantageous than slow, heavy-load resistances exercises.

PRACTICAL APPLICATIONS

Complex training theory has emerged in the last decade as a method commonly advocated for developing optimal muscular power in athletes (5,13). Our study provides preliminary support that pairing a hang clean, in comparison to a back squat, with CMJs was more effective in maintaining performance across sets. We also observed substituting hang cleans with an initial set of CMJs, resulted in better maintenance of CMJ performance. Moreover, the CMJ performance was most consistent when paired with hang cleans (Table 1), regardless of any potential biological differences among our 19 subjects. Thus, complex training may be preferred to training exclusively in one mode of exercise (i.e., resistance and plyometric exercise in separate sessions). Based on our findings, it seems plausible that pairing weightlifting techniques, as opposed to slow, heavy-load lifts, is a more preferred option for complex training. Our findings may not generalize to men or individuals of difference training statuses. Thus, research across multiple sets of paired exercises and with different populations is warranted.

REFERENCES

1. Cabrera, CA, Morales, J, Greer, F, and Pettitt, RW. Exercise bouts at three different intensities fail to potentiate concentric power. *Int J Exerc Sci* 2: 38–47, 2009.
2. Chiu, LZ, Fry, AC, Weiss, LW, Schilling, BK, Brown, LE, and Smith, SL. Postactivation potentiation response in athletic and recreationally trained individuals. *J Strength Cond Res* 17: 671–677, 2003.
3. Cormack, SJ, Newton, RU, McGuigan, MR, and Doyle, TLA. Reliability of measures obtained during single and repeated countermovement jumps. *Int J Sports Physiol Perf* 3: 131–144, 2008.
4. Henneman, E, Clamann, HP, Gillies, JD, and Skinner, RD. Rank order of motoneurons within a pool: law of combination. *J Neurophysiol* 37: 1338–1349, 1974.
5. Hodgson, M, Docherty, D, and Robbins, D. Post-activation potentiation: underlying physiology and implications for motor performance. *Sports Med* 35: 585–595, 2005.
6. Hopkins, WG. Measures of reliability in sports medicine and science. *Sports Med* 30: 1–15, 2000.
7. Hufschmidt, HJ. Wird durch muskeltvibration eine eigenreflexreihe erzeugt? *Pluegers Archiv European Journal of Physiology* 267: 508–516, 1958.
8. Jensen, RL, and Ebben, WP. Kinetic analysis of complex training rest interval effect on vertical jump performance. *J Strength Cond Res* 17: 345–349, 2003.
9. Kraemer, WJ, and Ratamess, NA. Fundamentals of resistance training: progression and exercise prescription. *Med Sci Sports Exerc* 36: 674–688, 2004.
10. MacIntosh, BR. Role of calcium sensitivity modulation in skeletal muscle performance. *News Physiol Sci* 18: 222–225, 2003.

11. McCann, MR, and Flanagan, SP. The effects of exercise selection and rest interval on postactivation potentiation of vertical jump performance. *J Strength Cond Res* 24: 1285–1291.
12. Palmer, BM, and Moore, RL. Myosin light chain phosphorylation and tension potentiation in mouse skeletal muscle. *Am J Physiol* 257: C1012–C1019, 1989.
13. Robbins, DW. Postactivation potentiation and its practical applicability: A brief review. *J Strength Cond Res* 19: 453–458, 2005.
14. Sale, DG. Postactivation potentiation: role in human performance. *Sport Sci Rev* 30: 138–143, 2002.
15. Sale, DG. Postactivation potentiation: role in performance. *J Sports Med* 38: 386–387, 2004.
16. Smith, CJ, and Fry, AC. Effects of a ten-second maximum voluntary contraction on regulatory myosin light-chain phosphorylation and dynamic performance measures. *J Strength Cond Res* 21: 73–76, 2007.
17. Verkhoshansky, Y. Perspectives in the improvement of speed-strength preparation for jumpers. *Track Field* 9: 11–12, 1966.
18. Weber, KR, Brown, LE, Coburn, JW, and Zinder, SM. Acute effects of heavy-load squats on consecutive squat jump performance. *J Strength Cond Res* 22: 726–730, 2008.